

Target DTE, Moneyness, and Calls vs Puts for Single-Leg Delta-Hedged SPY Options

Strategy research · L/S Gamma Desk, QUANTT

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Question: what target DTE (days to expiry) and moneyness (how far out of the money, delta) should we use for single-leg delta-hedged SPY options in a long/short gamma VRP strategy; and should long gamma use calls or puts and short gamma calls or puts

Method: Searched all 27 papers in `docs/papers/` (page-tagged index) + web search + the desk's own IBKR chain snapshot (2026-09-29). Quotes verified against the cited PDF pages. Page numbers are PDF pages; printed page numbers are given where they differ.

Related research: Volatility surface · Leverage effect · Size and exit rules

Desk decisions already made (PM, 2026-10-01): VRP band as v1 (long gamma if forecast RV – IV > +5%, short gamma if < –2%, else flat); single-leg calls and puts only to start; daily delta hedging.

1 Short answer

Enter **at the money (about 50 delta)**, at the **Friday expiry closest to 30 DTE**, within a 21–45 DTE window.

Use **long ATM calls for long gamma** and **short ATM puts for short gamma**:

- **Why ATM:** it carries the most gamma and vega per contract, and it's where S&P delta-hedged option losses, i.e. the VRP, are largest per option.
- **Why ~30 DTE:** Sepp finds the delta-hedged Sharpe ratio peaks at 1–2 month maturities.
- **Why calls for long, puts for short:** once delta-hedged, an ATM call and an ATM put are economically almost identical. The choice is operational. Short puts avoid early assignment around SPY's ex-dividend dates, and long calls avoid paying the put skew.

Selling *out-of-the-money* puts would collect more premium (puts are 3–4 vol points richer), but that's a later variant with more crash risk and less gamma to scalp.

2 What the papers say

2.1 Maturity: 1–2 months is the sweet spot

- **Sharpe peaks at 1–2 months:**

“We observe that the optimal Sharpe ratio peaks for options with maturities of one and two month and declines as maturity increases because expected transaction costs increase.” (Sepp 2013, *When You Hedge Discretely*, p. 17)

- **Short maturities need frequent hedging:**

“The optimal re-hedging period increases for longer maturities from as low as hedging once per 3 days for short-term maturities (up to 2 month) to 7–8 days for longer-term maturities (above 2 years).” (Sepp, p. 17)

So daily hedging is more frequent than Sepp’s optimum for 1–2 month options. That’s fine, but costs should be watched.

- **Practice converges on about one month:**

“at-the-money call options on S&P 500 with maturity of approximately 30 days” (Bracha, Sakowski & Michaków 2025, *DRL for ATM S&P 500 Options Hedging*, p. 6, printed p. 5)

“VRC buys the nearest monthly expiry ATM straddle” (Verma 2026, *PRISM*, p. 35)

- **Very short dates dominate volume but behave differently:**

“In 2022, more than 60% of the trading volume of S&P 500 index options was in options expiring within 7 calendar days.” (Chong & Todorov 2024, *Volatility of Volatility and Leverage Effect from Options*, p. 2)

There’s plenty of liquidity there, but gamma near expiry swings too fast for once-a-day hedging (inference).

2.2 Moneyness: at the money

- **Most gamma and vega per contract:**

“At-the-money options are efficient hedging instruments because they have a large gamma and vega compared with similar maturity options that are significantly in- or out-of-the-money.” (Cao, Chen, Farghadani, Hull, Poulos, Wang & Yuan 2022, *Gamma and Vega Hedging Using Deep Distributional RL*, p. 3)

“Whereas short maturity at-the-money options are most useful for hedging gamma, longer maturity options work better for vega.” (same, p. 17)

- **Standard practice in the RL papers:**

“the strategy selects the nearest-to-the-money (ATM) straddle within a predefined maturity range as the underlying instrument.” (Chen, Cai, Qin & An, *OPHR*, p. 26)

- **The theory fits ATM best:**

“Our current framework is best suited to analyzing at-the-money options for which the impact of the skew is limited.” (Sepp, p. 19)

- **ATM needs the most hedging:**

“Thus, for at-the-money options with high gamma, the rebalancing is expected to be more frequent, while, for out-of-the money options with low gamma, the re-hedging is only applied for big moves in the spot price.” (Sepp, p. 8)

- **ATM shorts take jumps hardest:**

“We note that a jump leads to a large realized loss in a short option position only if the option is near at-the-money.” (Sepp, p. 18)

2.3 Where the VRP is largest (web, Bakshi & Kapadia 2003)

The classic study of delta-hedged S&P 500 options (1988–1995, 14–60 day maturities) finds:

- “the delta-hedged strategy underperforms zero”
- “the documented underperformance is less for options away from the money”
- for ATM calls, the loss “amounts to 8% of the option value”
- the result is robust “to the inclusion of put options”

So **selling ATM collects the most premium per option**, and **buying ATM pays the most**. Long gamma should only be bought when the RV forecast clearly beats IV, which is what the +5% band enforces.

2.4 Calls vs puts

- **Option buyers lose most on puts, especially out of the money.** Sepp cites Broadie, Chernov & Johannes:

“the average monthly return (for long position in a put) is -30% for at-the-money puts and -57% for 6% out-of-the-money puts.” (Sepp, p. 3)

These are unhedged long-put returns, but they show where the richest premium sits.

- **The market prices in more crash risk than actually happens:**

“a stronger (more negative) leverage correlation under Q than under P.” (Hansen, Huang, Tong & Wang 2021, *Realized GARCH, CBOE VIX, and the VRP*, p. 3)

Q is the risk-neutral world implied by option prices; P is the real world. This gap is why OTM puts carry the skew premium (see the leverage report).

- **After delta hedging, call vs put at the same strike is mostly equivalent** (put-call parity; see docs/education/Forward_and_reverse_scalping.md, section 6). The real differences:
 - **Early exercise:** short ITM SPY calls can be assigned before ex-dividend dates.
 - **Skew:** matters for OTM strikes, little at ATM.
 - **Liquidity**, and the initial hedge size, which is about 50 shares per contract at ATM either way.

2.5 The desk’s own data (IBKR chain, 2026-09-29, spot 765.05)

DTE	ATM IV	25 Δ put IV	25 Δ call IV	25 Δ skew	10 Δ put IV	ATM gamma
17	12.6%	14.8%	11.4%	3.3 pts	17.7%	0.019
24	12.6%	14.9%	11.3%	3.5 pts	18.2%	0.016
31	13.2%	15.6%	11.8%	3.8 pts	19.4%	0.014
38	13.4%	16.1%	12.1%	4.1 pts	19.9%	0.012
52	13.6%	16.4%	12.1%	4.3 pts	20.6%	0.010

- The put skew is clear and grows with maturity.
- ATM gamma per contract roughly halves from 17 to 52 DTE.

3 How the papers connect

Source	Agrees with	Key contribution
Sepp 2013	Bakshi & Kapadia	Sharpe peaks at 1–2 month maturities; framework suited to ATM
Cao et al. 2022	OPHR, Bracha	ATM gives the most gamma and vega
Bakshi & Kapadia 2003 (web)	Sepp, Hansen	Delta-hedged losses (the VRP) are largest at the money
Hansen et al. 2021	Leverage report	Crash risk priced higher than it occurs, which drives put skew
OPHR, Bracha, PRISM	each other	Practice: nearest-ATM, about 1-month options

- **All sources point to ATM and roughly one-month options** for a delta-hedged VRP trade.
- **The only pull away from ATM is the skew premium** in OTM puts. It’s richer, but it comes with more crash exposure and less gamma.

4 Relevance to the LS Gamma desk

These are suggestions; the PM sets the final numbers:

- **Entry expiry:** the Friday expiry closest to 30 DTE within 21–45 DTE. The pipeline already pulls Friday expiries out to 60 DTE.
- **Strike:** the listed strike closest to spot, which the pipeline’s `atm_iv` logic already finds.
- **Long gamma** (forecast $RV - IV > +5\%$): `LG_LongCall`, ATM.
- **Short gamma** (forecast $RV - IV < -2\%$): `SG_ShortPut`, ATM.
- **IV for the signal:** use the ATM IV at the chosen expiry, not VIX, so the signal matches the option actually traded.
- **Later variant:** a short 25-delta put to harvest the skew, once the ATM version is running and measured.

5 Gaps and open questions

- **No paper tests single-leg, daily-hedged SPY options** with these exact choices. Bakshi & Kapadia’s data is 1988–1995, before weekly and daily expiries existed.
- **Daily hedging vs Sepp’s optimum.** Daily is more frequent than his roughly 3-day optimum, so watch commissions.
- **Dividend dates:** check SPY’s ex-dividend calendar if short calls are ever used.

6 Sources

Repo papers used, with pages cited:

- `When-You-Hedge-Discretely-Optimization-of-Sharpe-Ratio-for-Delta-Hedging-Strategy-under-Discrete-Hedging-and-Transaction-Costs.pdf`: Sepp 2013 (pp. 3, 8, 17, 18, 19)

- `Gamma-and-Vega-Hedging-Using-Deep-Distributional-Reinforcement-Learning.pdf`: Cao et al. (pp. 3, 17)
- `OPHR-Mastering-Volatility-Trading-with-Multi-Agent-Deep-Reinforcement-Learning.pdf` (p. 26)
- `Application-of-Deep-Reinforcement-Learning-to-At-the-Money-S&P-500-Options-Hedging.pdf`: Bracha et al. (p. 6)
- `PRISM-A-Probabilistic-Regime-Integrated-Scalping-Model-for-Derivatives-Arbitrage.pdf` (p. 35)
- `Volatility-of-Volatility-and-Leverage-Effect-from-Options.pdf`: Chong & Todorov (p. 2)
- `Realized-GARCH,-CBOE-VIX,-and-the-Volatility-Risk-Premium.pdf`: Hansen et al. (p. 3)

Checked, not relevant to DTE/moneyness: the forecasting papers, the agent and LLM papers, `From-Deep-Learning-to-LLMs-A-Survey-of-AI-in-Quantitative-Investment.pdf` (an AI-in-quant survey), and the remaining deep hedging papers. They use ATM or about 30-day options without studying the choice.

Web sources (outside the repo):

- Bakshi & Kapadia 2003, *Delta-Hedged Gains and the Negative Market Volatility Risk Premium* (RFS), JSTOR

Suggested papers to add to docs/papers/:

1. **Bakshi & Kapadia 2003:** the core evidence on delta-hedged option returns by moneyness and maturity.
2. **Broadie, Chernov & Johannes 2009, *Understanding Index Option Returns*:** the source of Sepp's put-return figures. Found only through Sepp's citation.